

# Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

**Project:** Quarter-Car Suspension Modeling and Simulation with Simscape Multibody

**Team Name:** QuarterCar7

**Instructor:** Dr. Esperanza Linares Guerrero & Dr. Harish Chinatakunta

**Program/Class:** MathWorks Classroom Challenge Projects

## Members and Roles

1. Ryan Law
2. Collin Wong
3. Noel Chen
4.
5.

## 1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

## 2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

## 3) Communication Norms

- Primary channel: Discord (e.g. Slack)
- Secondary channel: Email (e.g. E-mail for summaries/decisions)
- Response time: 24 hours on weekdays and 48 Hours on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: \_\_\_\_\_ (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom)  
PM logs major decisions, assumptions, and task updates on
- Decision records: shared README file (e.g. PM logs major decisions in a running document)

## 4) Collaboration & Tools

- Version control: shared GitHub repository (e.g. one shared GitHub repo or Google Drive folder)

- Data management: All datasets, parameters, and assumptions documented in README.md (e.g. README.md)
- File naming & style: quarter\_car\_model\_v1.slx (e.g. Snake\_case, descriptive names)  
Older versions will be saved in an
- Backups: archive folder (e.g. PM tags old versions according to numbered system)

## 5) Quality Standards (Definition of 'Done')

Work is considered complete when the project requirements are met, the Simscape Multibody model runs without errors, the MATLAB scripts reproduce the required plots, assumptions and units are documented, and at least one teammate other than the original creator has tested the file. Final figures, calculations, and conclusions must match the project description and be clearly explained in the report. The final repository or folder must be organized so another person could open the README and understand how to run the project.

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

## 6) Timeline & Milestones

- Week 1: Confirm roles, parameters, and create model plan.
- Week 2: Build baseline Simscape quarter-car model.
- Week 3: Test model, check units, and validate results.
- Week 4: Finalize plots, report, slides, and files.

(e.g.

Week 1: Problem framing, literature/background review, dataset/parameters,  
Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

Week 3: Validation and quality assurance

Week 4: Report, slides, presentation rehearsal, repository cleanup)

## 7) Decision-Making & Conflict Resolution

- Default: Consensus after discussion, PM facilitates (e.g. consensus after discussion; PM facilitates)
- If blocked >48 hrs: Majority Vote (e.g. Time-limited debate, majority vote, PM documents decision)  
Evidence based comparison and
- Technical disagreements: instructor/mentor guidance before voting. (e.g. Evidence-based comparison before vote)
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

## 8) Workload, Accountability & Attendance

- Task board: To-Do and due dates documented in README (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)  
Missed deadlines require a recovery plan within 24 hours.
- Accountability: Repeated misses may lead to role reassignments. (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)  
If a teammate will miss a meeting, they should give
- Attendance: 24-hour notice when possible (e.g. If you'll miss a meeting, give 24-hour notice and share updates asynchronously.)

## 10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

## 11) Deliverables Checklist

Working Simscape Multibody quarter-car suspension model

MATLAB scripts needed to run simulations and generate plots

Clear README with instructions, assumptions, units, and parameter values

Hand calculations or validation notes linked to simulation results

Plots of road input, sprung mass motion, unsprung mass motion, and suspension response

Final report with problem description, model setup, assumptions, results, validation, and conclusion

Slides or presentation materials if required

Organized shared folder or repository with final files clearly labeled

Final quality check confirming that the model and scripts run correctly

\_\_\_\_\_  
\_\_\_\_\_

( e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

## 12) Agreement & Signatures

Name & Signature: Ryan Law Date: \_\_\_\_\_

07/08/2026

Name & Signature: Collin Wong Date: \_\_\_\_\_

07/08/2026

Name & Signature: Noel Chen Date: \_\_\_\_\_

07/08/2026

Name & Signature: \_\_\_\_\_ Date:

07/08/2026